

• Independent Systems

A system of equations is said to be independent if all the equations describe distinct constraints on the variables. Such a system typically has a unique solution, where no equation can be derived as a linear combination of the others.

This means:

1. Unique Constraints: Each equation in an independent system represents a distinct constraint on the variables. None of the equations can be derived from a combination of the others.
2. Unique Solution: Because each equation is uniquely contributing to the constraints, an independent system typically has a single solution that satisfies all equations simultaneously. This single solution represents the intersection point (or points, in higher dimensions) where all equations meet.
3. Example of an Independent System: Consider a simple system:

$$\begin{cases} x + y = 5 \\ 2x - y = 1 \end{cases}$$

Here, both equations contribute unique constraints, and solving them will lead to one specific solution, making it an independent system.

• Dependent System

A system is considered dependent when one or more equations are linearly dependent on others. This means that one equation can be written as a linear combination of the others, leading to an infinite number of solutions (if the system is consistent) or no solution (if inconsistent).

This means:

1. Redundant Information: In a dependent system, one or more equations can be formed by a linear combination of the others. This redundancy means that not all equations contribute unique constraints.
2. Infinite Solutions or No Solutions:
 - If the system is consistent (the equations do not contradict each other), a dependent system will have infinitely many solutions, as there are “extra” degrees of freedom due to the dependent equations.
 - If the system is inconsistent (e.g., parallel but non-intersecting planes in three dimensions), it will have no solution.

3. Example of a Dependent System:

$$\begin{cases} x + y = 2 \\ 2x + 2y = 4 \end{cases}$$

In this case, the second equation is simply twice the first, so both equations represent the same line. Since they don't offer unique constraints, this system is dependent and has infinitely many solutions along that line.

Problems

a) Solve the system:

$$\begin{cases} x + y + z = 5 \\ 5x + 3y + 2z = 0 \\ y - z = 0 \end{cases}$$

1. Simplify the System

From the third equation, is possible to express z in terms of y :

$$y - z = 0 \Rightarrow z = y.$$

Substitute $z = y$ into the first and second equations:

- $x + y + y = 5 \Rightarrow x + 2y = 5.$
- $5x + 3y + 2y = 0 \Rightarrow 5x + 5y = 0.$

Now, there is the following simplified system:

$$\begin{cases} x + 2y = 5 \\ 5x + 5y = 0 \end{cases}$$

2. Solve for x and y

Divide the second equation by 5 to simplify:

$$x + y = 0.$$

Now there are:

$$\begin{cases} x + 2y = 5 \\ x + y = 0 \end{cases}$$

Subtract the second equation from the first equation to eliminate x :

$$(x + 2y) - (x + y) = 5 - 0 \Rightarrow y = 5.$$

Now that is known that $y = 5$, substitute this value into the second equation:

$$x + 5 = 0 \Rightarrow x = -5.$$

3. Find z using $z = y$

Since $z = y$:

$$z = 5.$$

The solution to the system is:

$$(x, y, z) = (-5, 5, 5).$$

This means the equations in this system intersect at exactly one point in 3-dimensional space, so they are linearly independent and have a unique solution.

b) Solve this system of linear equations:

$$\begin{cases} x_1 - x_2 - 3x_3 + 4x_4 = 1 \\ x_1 + x_2 + x_3 + 2x_4 = -1 \\ -x_2 - 2x_3 + x_4 = 1 \\ x_1 + 2x_2 + 3x_3 + x_4 = -2 \end{cases}$$

1. Eliminate x_1 from Equations 2 and 4. Starting with the first equation, use it to eliminate x_1 from the other equations.

From Equation 1:

$$x_1 = x_2 + 3x_3 - 4x_4 + 1$$

Substitute into Equation 2:

$$(x_2 + 3x_3 - 4x_4 + 1) + x_2 + x_3 + 2x_4 = -1$$

Combine like terms:

$$2x_2 + 4x_3 - 2x_4 = -2$$

Simplify:

$$x_2 + 2x_3 - x_4 = -1 \quad (\text{Equation 5})$$

Substitute into Equation 4:

$$(x_2 + 3x_3 - 4x_4 + 1) + 2x_2 + 3x_3 + x_4 = -2$$

Combine like terms:

$$3x_2 + 6x_3 - 3x_4 = -3$$

Simplify:

$$x_2 + 2x_3 - x_4 = -1 \quad (\text{Equation 6})$$

2. Simplify the System. Now there is a simplified system with three equations:

$$\begin{cases} x_2 + 2x_3 - x_4 = -1 & (\text{from Equation 5}) \\ -x_2 - 2x_3 + x_4 = 1 & (\text{from Equation 3}) \\ x_2 + 2x_3 - x_4 = -1 & (\text{from Equation 6}) \end{cases}$$

Notice that Equations 5 and 6 are identical, so was left with just two unique equations:

$$\begin{cases} x_2 + 2x_3 - x_4 = -1 \\ -x_2 - 2x_3 + x_4 = 1 \end{cases}$$

3. Add the Equations to Eliminate x_2 and x_4 .

Add these two equations:

$$(x_2 + 2x_3 - x_4) + (-x_2 - 2x_3 + x_4) = -1 + 1$$

This simplifies to:

$$0 = 0$$

This identity suggests that the equations are dependent, meaning there are infinitely many solutions. To find the general solution, is possible to express some variables in terms of others.

4. Express x_2 in Terms of x_3 and x_4 .

From Equation 5:

$$x_2 + 2x_3 - x_4 = -1$$

Solving for x_2 :

$$x_2 = -1 - 2x_3 + x_4$$

5. Express x_1 in Terms of x_3 and x_4 .

Now, using the expression for x_1 from Step 1:

$$x_1 = x_2 + 3x_3 - 4x_4 + 1$$

Substitute $x_2 = -1 - 2x_3 + x_4$:

$$x_1 = (-1 - 2x_3 + x_4) + 3x_3 - 4x_4 + 1$$

Simplify:

$$x_1 = x_3 - 3x_4$$

6. The solution in terms of free variables x_3 and x_4 is:

$$\begin{cases} x_1 = x_3 - 3x_4 \\ x_2 = -1 - 2x_3 + x_4 \\ x_3 = x_3 \\ x_4 = x_4 \end{cases}$$

The system is dependent since has many solutions depending on two parameters:

$$(x_3 - 3x_4, -1 - 2x_3 + x_4, x_3, x_4), \forall x_3, x_4 \in \mathbb{R}$$

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.